

UI/UX & Theme System

The reusable design + research library for every Alongside-family app. Drop this into a new build chat and Claude has the philosophy, the laws, the ADHD/retention evidence, the component patterns, and a library of ready-to-apply themes — so we never re-run this research per app.

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§0 · Purpose & how to use this document

What this is

A single, self-contained reference that captures **how our apps should look and feel** and **why** — the UX laws, the ADHD & retention psychology, the component patterns, and a curated library of themes. The visual research is expensive to do well and easy to get wrong; this file makes it a one-time cost.

How to use it in a new app chat

1. Upload this PDF and say what the app is (e.g. "a habit tracker", "a client CRM", "a plant monitor").
2. Pick a theme by number from **§6** (or ask for a recommendation — use the matrix in **§7**).
3. Claude applies the **shared component layer (§5)** + the chosen **theme token block (§6)**, follows the **philosophy (§1)**, the **laws (§2)**, the **ADHD rules (§3)**, and the **card logic (§4)**, and ships against the **quality floor (§8)**.

What the end result should be

A distinctive, calm, ADHD-aware, accessible interface that doesn't read as a generic AI default — built in a fraction of the time because the rules, laws, logic, and psychology are already settled here. **The themes are not separate codebases.** They are **token sets** over one shared component layer: swapping a theme = swapping ~15 CSS variables, not rebuilding screens.

Document rules (continuity)

- **One file, append-only.** Substantive changes get a dated entry in the changelog (§10) and a version bump. Edit core sections in place only when a decision genuinely changes.
- **Versioning** MAJOR.MINOR.PATCH : MINOR = new theme or new pattern; PATCH = fixes/tweaks; MAJOR = a token-contract or structural break.
- **Sources drive behavior.** Any claim that sets a rule cites a source (§9). Contested claims are flagged, not asserted.
- **Honesty over polish.** Where evidence is weak, we say so and avoid over-engineering mechanics on top of it.

§1 · Design philosophy (non-negotiables)

These hold across every app and every theme. They are the spine; the themes are the skin.

- **Calm, glanceable, never shaming.** The interface lowers cognitive load and never scolds. A missed day is met with "water when you can," never "you forgot."
- **One focal thing per screen.** A single, visually-isolated "do this next" / "needs you now" element (Von Restorff). If everything is emphasized, nothing is.
- **Honest reward, never fake progress.** Satisfying completion and real milestones — **no points, streaks, or manufactured urgency.** Shame and manipulation churn ADHD users.
- **Externalize time and state.** Make "overdue", "last updated", "days since", and "now" explicit and visible. The brain that can't feel time gets it drawn on screen.
- **Aesthetic-usability + warmth, not flashy.** Beautiful and legible. A space that feels good gets reopened; beauty must serve function, never replace it.
- **Status meaning is constant.** Good/watch/bad colors never change meaning between light and dark. Tone flips; semantics don't.
- **Avoid the AI-default look.** Steer around the three tells: (1) cream + elegant-serif + terracotta accent; (2) near-black + one acid-green/vermillion accent; (3) hairline broadsheet columns. Ground the look in the app's real subject instead. Spend boldness in *one* signature element and keep the rest quiet.
- **Accessibility is a floor, not a finish.** See §8.

§2 · UX laws → the build move

Each law, what it is, and the concrete move it justifies in our apps. Links in §9.

LAW	WHAT IT MEANS	OUR MOVE
Hick's Law	Decision time grows with the number/complexity of choices.	One focal action per screen; segmented add (library / single / brain-dump); progressive disclosure of the rest.
Von Restorff	The one distinct element is seen and acted on first.	The "needs you now" card / Now-Bar status pill is the single isolated focal element.
Fitts's Law	Target acquisition time depends on size + distance.	Primary actions are large; min 44×44px touch targets; thumb-reachable bottom nav.
Doherty Threshold	Engagement holds when system response is < 400 ms.	Snappy animation (bars ~300–400 ms, gauge sweep ~500–600 ms); animate transform/opacity only; honor reduced-motion.
Goal-Gradient / Zeigarnik	Motivation rises near a goal; open loops nag.	Visible, satisfying completion. Caveat Zeigarnik & endowed-progress have mixed replication — use the honest "completion feels good" idea, don't build mechanics on contested effects.
Aesthetic-Usability	Attractive UIs are perceived as more usable; users forgive minor flaws.	Invest in a coherent, warm visual identity — but only where it supports function.
Cognitive Load	Cut extraneous load; protect working memory.	Chunk into zones, color-code salience, hide secondary detail behind ≤2 levels of disclosure.
Progressive Disclosure	Show primary first; defer the rest; keep ≤2 levels.	One reusable collapsible for galleries, history, money detail; calm default view.
Miller's Law	~7±2 items in working memory (recall) .	See the correction in §3 — this is not a cap on visible on-screen items.
Jakob's Law	Users expect your app to work like the others they know.	Conventional nav, icons, and gestures; innovate on the one thing that differentiates, not on basics.
Few (data-viz)	A bare value is weak; a dial needs a target + qualitative bands.	Every gauge ships with a target marker and good/watch/bad color bands (the bands <i>are</i> the thresholds).

§3 · ADHD & app-retention research

Why the philosophy is shaped the way it is. These findings set rules; sources in §9.

- **R1 — The ADHD brain needs external scaffolding.** Dysregulated dopamine/norepinephrine in the prefrontal cortex impairs sustained attention on low-reward tasks. Tools provide the executive function the brain can't self-generate: reward loops, externalized time/structure, accountability.
- **R2 — Retention failure modes (the design brief).** "Complicated layouts or too many menu layers lead to abandonment." When mainstream tools fail, ADHD users **blame themselves**, reinforcing shame. So: reduce decisions, externalize memory, remove friction, and **never punish inconsistent use**.
- **R4 — Implementation intentions.** Specifying *when/where/how* ("if situation Y, then behavior X") has a medium-to-large effect on follow-through (Gollwitzer & Sheeran 2006, $d = .65$ across 94 tests). Built as: concrete "do this next", one-tap relative dates, and "comes after" dependencies (if-then chains).
- **R5 — Time blindness (Barkley).** Future deadlines don't feel real until they're emergencies. Countermeasure: draw the clock — explicit due dates, bold "overdue / Nd ago", a "today" marker, and IoT freshness ("updated 3m ago"). A green reading on a sensor that died hours ago is a dangerous lie; treat **stale/offline as distinct from bad**.
- **R6 — Timeboxing / Pomodoro.** Externalizing time ("work until the alarm") sidesteps the need to judge time internally. Built as an optional focus timer (15/25/50), framed honestly (below).

⚠ Correction 1 — Body doubling is over-sold in older notes.

The claim that body doubling "cut procrastination ~30%" is **not supported**.

The evidence base is early, mixed, and largely anecdotal / master's-thesis level; controlled studies have not shown conclusive effects, and reviews note some neurodivergent discomfort with social presence. One controlled VR study found in-person body-double task completion ~17% vs ~12% with none — real but **modest**

. Rule: keep the optional co-working/"tending session" timer (low-risk, some find it helpful) but frame it honestly ("a quiet co-working timer some people with ADHD find helpful"). The firmer mechanism is Pomodoro/timeboxing time-externalization, not body doubling.

△ Correction 2 — Miller's "7±2" is not an on-screen element limit.

7±2 is about

working-memory recall

, not how many items may be

visible

. Visible items use

recognition

, not recall — the user can just look. A well-organized dashboard can show

8–12 glanceable metrics

if they're chunked into zones, color-coded, and ranked (one hero, the rest subordinate). Even Laws of UX warns not to use the "magical number seven" to justify design limits. Rule: never cap a dashboard at 7 tiles to "respect Miller." Reduce *extraneous* load via hierarchy and chunking instead.

Guardrails (carry into every app)

No points / streaks / fake progress. Gentle, non-judgmental microcopy for lapses and for over-correction alike ("looks a little wet — ease off watering", never "you overwatered"). Honest milestone toasts only (reached flowering, first harvest). Aesthetic warmth, not gamification.

\$4 · When to use a card (and when not)

The most common mistake is treating cards as on/off — boxing everything, or boxing nothing. Cards are a tool for a **specific job**: visually separating a discrete, independent, scannable unit. Over-carding adds borders and noise and flattens hierarchy (if every block is a card, the focal one can't stand out); zero-carding loses grouping and tap affordance.

USE A CARD WHEN...	DON'T USE A CARD WHEN...
It's a discrete, independent object (a plant in a grid, a task row that opens a detail).	It's the immersive hero — full-bleed media (the live photo/video) carries more by bleeding edge-to-edge.
It's the single focal action that should be isolated + elevated (the "needs you now" alert).	It's chrome — the nav/top bar is structure, not content; keep it integrated.
It groups related data into one unit (soil + VPD + air = "this plant now").	It's a single primary reading or headline — a big number breathes better on the background.
You need elevation to signal an interactive surface distinct from the page.	It's a continuous reading flow — a hairline divider or whitespace groups more elegantly than a box.

Worked example (the plant screen we converged on): full-bleed photo/video (no card) → one **focal alert card** → one **grouped-vitals card** (soil leads, VPD/air below) → integrated nav (no card). Result: ~2 purposeful cards, not 4, not 0.

Component catalogue (theme-agnostic)

- **Full-bleed media hero** — the live feed/photo edge-to-edge; name overlaid on a scrim; an "offline · last photo" state (never a blank panel — fall back to the last image).
- **Focal-alert card** — Von Restorff isolation; accent edge; one-tap action; no-shame copy.
- **Grouped-vitals card** — related live readings; one hero metric + supporting metrics with a hairline between.
- **Band / tide meter** — a value on a track with a marker; for status, a good/watch/bad zoned band (Few).
- **Gauge done right** — radial dial with target tick + qualitative bands; never a bare arc.
- **Now-Bar nav** — 4 destinations + 1 living center status pill that always shows the single most urgent thing, color-coded; on desktop becomes a left rail + a top "needs attention" banner.
- **Freshness / offline indicator** — fresh / stale / offline shown per sensor; prevents "green but dead".
- **Progressive-disclosure collapsible** — one reusable section; secondary detail folded by default.

\$5 · Token architecture + the shared layer

This is the engine. Components reference variables; a theme is a set of variables. Build the shell once, then drop in any theme block from \$6.

The token contract

TOKEN	MEANING
--bg	Page background (gradient or solid).

<code>--bgfade</code>	The solid bottom color of <code>--bg</code> — used to fade media into the page and behind the nav.
<code>--ink</code> / <code>--soft</code>	Primary text / secondary text.
<code>--metaink</code>	Text that sits over media (always light).
<code>--accent</code> / <code>--accent2</code>	Brand accent + its partner for gradients.
<code>--good</code> / <code>--watch</code> / <code>--bad</code>	Status. Meaning is constant across themes.
<code>--rule</code> / <code>--ruleV</code>	Horizontal / vertical hairline (often a subtle gradient).
<code>--track</code>	The empty portion of a meter.
<code>--card</code> / <code>--cardln</code>	Card surface / card hairline border.
<code>--display</code> / <code>--body</code> / <code>--data</code>	Display font / body font / mono-numeric font.

Shared component CSS (uses the tokens above)

```
.screen{background:var(--bg); color:var(--ink); font-family:var(--body); position:relative; overflow:hidden; display:flex; flex-direction:column}

/* full-bleed media hero */
.feed{position:relative; width:100%; aspect-ratio:4/5}
.feed .photo, .feed video, .feed img{position:absolute; inset:0; width:100%; height:100%; object-fit:cover}
.feed .tint{position:absolute; inset:0; background:linear-gradient(180deg, rgba(3,14,18,.55) 0,
  rgba(3,14,18,0) 19%, rgba(3,14,18,0) 46%, rgba(3,14,18,.5) 74%, rgba(3,14,18,.92) 100%)}
.feed .fade{position:absolute; left:0; right:0; bottom:0; height:26%;
  background:linear-gradient(180deg, transparent, var(--bgfade))}
.feed .fmeta{position:absolute; left:0; right:0; bottom:0; padding:0 20px 18px; z-index:3}
.feed .fmeta h1{font-family:var(--display); font-weight:700; font-size:2.3rem; letter-spacing:-.025em;
  color:var(--metaink); text-shadow:0 2px 20px rgba(0,25,25,.5); margin:0}

/* card: used only where it earns it (see §4) */
.card{background:var(--card); border:1px solid var(--cardln); border-radius:18px;
  backdrop-filter:blur(15px); -webkit-backdrop-filter:blur(15px)}

/* focal alert (Von Restorff) */
.needs{padding:14px 15px; display:flex; align-items:center; gap:13px; position:relative; overflow:hidden}
.needs::before{content:""; position:absolute; left:0; top:0; bottom:0; width:4px; background:var(--bad)}
.needs .go{margin-left:auto; font-weight:700; color:var(--accent)}

/* grouped vitals */
.vitals{padding:16px}
.soil .num{font-family:var(--display); font-weight:700; font-size:3rem; letter-spacing:-.03em; color:var(--ink)}
.tide{position:relative; height:7px; border-radius:99px; background:var(--track); margin-top:14px}
.tide::after{content:""; position:absolute; inset:0; width:var(--p,60%); border-radius:99px;
  background:linear-gradient(90deg, var(--accent2), var(--accent))}
.rule{height:1px; margin:16px 0; background:var(--rule)}
.duo{display:grid; grid-template-columns:1fr auto 1fr; gap:16px}
.duo .vsep{width:1px; background:var(--ruleV)}

/* Now-Bar nav (chrome – integrated, not a card) */
.nowbar{margin-top:auto; display:flex; align-items:center; justify-content:space-between; gap:6px;
  padding:14px 18px; background:linear-gradient(180deg, transparent, var(--bgfade) 62%)}
.nowbar .pill{min-width:126px; padding:9px 14px; border-radius:99px; text-align:center;
  background:linear-gradient(95deg, var(--accent2), var(--accent))}

/* quality floor */
:focus-visible{outline:2.5px solid var(--accent); outline-offset:3px}
@media (prefers-reduced-motion:reduce){*{animation:none !important; transition:none !important}}
```

HTML skeleton

```
<div class="screen" data-theme="lagoon">
  <div class="statusbar">...</div>
  <div class="scroll">
    <div class="feed">
      <!-- full bleed: photo OR <video> -->
      <div class="photo">...</div><div class="tint"></div><div class="fade"></div>
      <div class="fmeta"><h1>Blue Dream</h1><div class="sp">Seedling · day 3</div></div>
    </div>
    <div class="body">
      <div class="needs card">...focal alert...</div>
      <div class="vitals card">...soil hero + duo(VPD|air)...</div>
    </div>
  </div>
  <nav class="nowbar">... 4 items + center .pill ...</nav>
</div>
```

Theming model (light ↔ dark)

Themes are **tonal poles of one system**. The toggle flips **tone**, **never structure**: same components, same layout, same card decisions; only the token block changes. Status colors stay constant so meaning never shifts. Many themes naturally pair (a light pole + a dark pole) — see §6 and §7.

§6 · The theme library

Nine options. 1–7 came from the plant-app exploration; 8–9 are added for range (professional / business apps). For each: concept, best-for, fonts, signature, and the token block to paste into `[data-theme="..."]`.

1 · Lagoon LIGHT

Concept: looking into clear shallow tropical water. Airy, weightless, calm. **Best for:** wellness, plants, daily-glance apps — the default light. **Fonts:** Bricolage Grotesque / Hanken Grotesk / Space Mono. **Signature:** warm-sand & coral against ocean turquoise; "horizon" hairlines shading coral → turquoise. Locked
palette — approved, do not drift.



```
--bg:radial-gradient(95% 42% at 82% -6%, rgba(255,201,128,.45), transparent 56%),
  linear-gradient(180deg,#d4f1ef 0%, #e8f1ea 44%, #f7e8cb 100%);
--bgfade:#f7e8cb; --ink:#07313a; --soft:#4e7672; --metaink:#f6fcfb;
--accent:#0fb1bf; --accent2:#37cdd5; --good:#0fb1bf; --watch:#e89a1e; --bad:#ff7150;
--rule:linear-gradient(90deg, rgba(255,113,80,0), rgba(255,113,80,.4) 20%,
  rgba(15,177,191,.4) 80%, rgba(15,177,191,0));
--ruleV:linear-gradient(180deg, transparent, rgba(7,49,58,.18), transparent);
--track:rgba(7,49,58,.12); --card:rgba(255,255,255,.5); --cardln:rgba(255,255,255,.82);
--display:"Bricolage Grotesque"; --body:"Hanken Grotesk"; --data:"Space Mono";
```

2 · Reef Night (Bioluminescent) DARK

Concept: deep ocean at night — electric cyan into violet, glows that emit light. Vibrant, high-end. **Best for:** the **dark pole of Lagoon**; anything wanting a premium, alive dark. **Signature:** cyan → violet glow on the tide bar, status pill, alert dot, active nav. Add `box-shadow` glows on those elements only.



```
--bg:radial-gradient(92% 40% at 16% -6%, rgba(60,210,255,.20), transparent 52%),
  radial-gradient(82% 38% at 92% 2%, rgba(168,92,255,.20), transparent 52%),
  radial-gradient(70% 32% at 50% 108%, rgba(47,245,224,.12), transparent 62%),
  linear-gradient(180deg,#04141f 0%, #030c15 52%, #01070d 100%);
--bgfade:#01070d; --ink:#eaf7f7; --soft:#84b3bd; --metaink:#f4ffff;
--accent:#2ff5e0; --accent2:#8b5cff; --good:#2ff5e0; --watch:#ffce4d; --bad:#ff5e8a;
--rule:linear-gradient(90deg, rgba(139,92,255,0), rgba(139,92,255,.55) 20%,
  rgba(47,245,224,.55) 80%, rgba(47,245,224,0));
--ruleV:linear-gradient(180deg, transparent, rgba(120,205,225,.22), transparent);
--track:rgba(140,210,220,.13); --card:rgba(110,205,230,.055); --cardln:rgba(120,220,240,.16);
/* glows: .pill, .tide::after, .needs.d, .nb.on get box-shadow / text-shadow in --accent + --accent2 */
```

3 · Vivarium DARK

Concept: a plant through greenhouse glass at dusk. Immersive, premium, cooler than Reef Night. **Best for:** photo/video-led dark UIs. **Signature:** frosted-glass cards over the media; a slow grow-light glow that breathes (`@keyframes`, reduced-motion guarded).



```
--bg:linear-gradient(180deg,#0d1512,#11201a); --bgfade:#0b120f;
--ink:#eaf3ec; --soft:#9fb6a6; --metaink:#eaf3ec;
--accent:#7be0b0; --accent2:#5fcf9e; --good:#6fd08a; --watch:#e7b964; --bad:#e08a74;
--rule:linear-gradient(90deg, transparent, rgba(123,224,176,.3), transparent);
--ruleV:linear-gradient(180deg, transparent, rgba(123,224,176,.22), transparent);
--track:rgba(207,238,222,.14); --card:rgba(207,238,222,.10); --cardln:rgba(207,238,222,.22);
--display:"Bricolage Grotesque"; --body:"Hanken Grotesk"; --data:"Space Mono";
/* signature: stronger blur on .card; a breathing radial glow layer behind .feed */
```

4 • Tiki DARK

Concept: jungle shade lit by tropical-flower color. Vivid, playful. **Best for:** a high-energy dark. **Signature:** hibiscus-pink + plumeria-yellow flashes over deep palm green. **Green caveat** our plant-app went green-free (Reef Night) by request; Tiki keeps its jungle green as a distinct option — swap the leaf-green for teal if green is unwanted.



```
--bg:linear-gradient(170deg,#0f3b33 0%, #0c302b 60%, #0a2622 100%); --bgfade:#0a2622;
--ink:#f0f7ee; --soft:#9fc2ad; --metaink:#ffffff;
--accent:#3fb8b0; --accent2:#ff5d8f; --good:#46c79a; --watch:#ffd24a; --bad:#ff5d8f;
--rule:linear-gradient(90deg, rgba(63,184,176,0), rgba(63,184,176,.4), rgba(255,93,143,.4), rgba(255,93,143,0));
--ruleV:linear-gradient(180deg, transparent, rgba(160,200,180,.2), transparent);
--track:rgba(255,255,255,.1); --card:rgba(255,255,255,.05); --cardln:rgba(255,255,255,.12);
/* signature: a tropical spectrum bar (aqua+plumeria+hibiscus) for the hero meter */
```

5 • Sunset Shore LIGHT

Concept: golden hour on the sand — peach into coral into a cool dusk edge. Warm, dreamy, unhurried. **Best for:** a warm light alternative to Lagoon. **Fonts:** Fraunces / Hanken / Space Mono. **Signature:** a glowing sun-ring gauge; a warm horizon wash at the top.



```
--bg:radial-gradient(70% 36% at 50% 6%, rgba(255,210,150,.7), transparent 62%),
  linear-gradient(180deg,#ffe9d2 0%, #ffd4be 42%, #f4c4ac 72%, #e7c1bf 100%);
--bgfade:#e7c1bf; --ink:#4a3b34; --soft:#937a6e; --metaink:#fff6ee;
--accent:#ff8a6b; --accent2:#f0a93a; --good:#cf8a3a; --watch:#e08a4a; --bad:#d8556a;
--rule:linear-gradient(90deg, transparent, rgba(255,138,107,.35), rgba(240,169,58,.35), transparent);
--ruleV:linear-gradient(180deg, transparent, rgba(74,59,52,.18), transparent);
--track:rgba(74,59,52,.12); --card:rgba(255,250,244,.62); --cardln:rgba(255,255,255,.7);
--display:"Fraunces"; --body:"Hanken Grotesk"; --data:"Space Mono";
```

6 • Herbarium LIGHT

Concept: a pressed-specimen field journal + brass laboratory instrument. Scholarly, tactile, analog-meets-digital. **Best for:** data-rich, reference-heavy apps. **Fonts:** Fraunces italic / Hanken / Space Mono. **Signature:** an engraved needle gauge (conic ticks + needle); letterpress small-caps labels; specimen-framed media.



```
--bg:linear-gradient(180deg,#e9e2d0,#e4ddcb); --bgfade:#e4ddcb;
--ink:#23271b; --soft:#6b6450; --metaink:#fbf7ec;
--accent:#9a7b3f; --accent2:#43603e; --good:#4e8c57; --watch:#b9852f; --bad:#a85741;
--rule:linear-gradient(90deg, transparent, #c9c0a8, transparent);
--ruleV:linear-gradient(180deg, transparent, #c9c0a8, transparent);
--track:rgba(35,39,27,.1); --card:#f3eee0; --cardln:#c9c0a8;
--display:"Fraunces"; --body:"Hanken Grotesk"; --data:"Space Mono";
/* signature: radial gauge - repeating-conic ticks + a rotated needle (rotate by value) */
```

7 • Spectrum DARK

Concept: a grow-room control panel on the LED light spectrum (violet → magenta → amber). Technical, data-forward — the maker/ESP32 identity. **Best for:** dashboards, dev tools. **Signature:** a spectrum status bar; mono numerics.

⚠ Spectrum grid fix.

The original used a full graph-paper grid background — it read as "horrendous" / cheap. **Rule:** drop the full grid. If you want a technical texture, use a

sparse edge grid

, a single baseline rule, or a faint dot at low opacity — never a dense graph-paper fill behind content.



```
--bg:radial-gradient(120% 60% at 50% -10%, #241338 0%, transparent 55%),
  linear-gradient(180deg,#140d1c,#100b16); /* NO graph-paper grid */
--bgfade:#100b16; --ink:#f2ecf6; --soft:#a99bb4; --metaink:#ffffff;
--accent:#8c6cf0; --accent2:#e05ca8; --good:#58c97e; --watch:#f2a93b; --bad:#f0698f;
--rule:linear-gradient(90deg, transparent, rgba(140,108,240,.45), rgba(224,92,168,.45), transparent);
--ruleV:linear-gradient(180deg, transparent, rgba(140,108,240,.3), transparent);
--track:#241b30; --card:rgba(255,255,255,.018); --cardln:#2a2233;
--display:"Space Grotesk"; --body:"Hanken Grotesk"; --data:"Space Mono";
```

8 • Atelier LIGHT ADDED FOR RANGE

Concept: editorial-minimal studio — cool paper, deep ink, one confident accent, generous whitespace. Deliberately **not** the cream-serif-terracotta default. **Best for:** professional / business apps (e.g. the Cockpit). **Fonts:** Bricolage Grotesque / Hanken / Space Mono. **Signature:** restraint — type and whitespace do the work; cards are minimal with a 1px border, no glass.



```
--bg:#f3f4f2; --bgfade:#f3f4f2; --ink:#181a17; --soft:#676b64; --metaink:#ffffff;
--accent:#2f57d6; --accent2:#5b79e6; --good:#1f8a5a; --watch:#c08a1e; --bad:#c0492f;
--rule:linear-gradient(90deg, transparent, rgba(24,26,23,.14), transparent);
--ruleV:linear-gradient(180deg, transparent, rgba(24,26,23,.14), transparent);
--track:rgba(24,26,23,.1); --card:#ffffff; --cardln:rgba(24,26,23,.1);
--display:"Bricolage Grotesque"; --body:"Hanken Grotesk"; --data:"Space Mono";
```

9 • Graphite DARK ADDED FOR RANGE

Concept: a refined neutral instrument — charcoal, warm-neutral text, one steel-blue accent. Sober, crisp, no glow. The **dark pole of Atelier**. Deliberately **not** the near-black + acid-green default. **Best for:** professional dark mode. **Signature:** precision and contrast, not color.



```
--bg:linear-gradient(180deg,#16181a,#121416); --bgfade:#121416;
--ink:#eceeec; --soft:#9aa09a; --metaink:#ffffff;
--accent:#4f9fd0; --accent2:#6fb6e0; --good:#5bbf8a; --watch:#d8a23a; --bad:#d4634f;
--rule:linear-gradient(90deg, transparent, rgba(180,190,190,.18), transparent);
--ruleV:linear-gradient(180deg, transparent, rgba(180,190,190,.18), transparent);
--track:rgba(180,190,190,.13); --card:rgba(255,255,255,.03); --cardln:rgba(255,255,255,.1);
--display:"Bricolage Grotesque"; --body:"Hanken Grotesk"; --data:"Space Mono";
```

§7 • Choosing a theme

IF THE APP IS...	LIGHT POLE	DARK POLE
Plant / wellness / daily-glance (calm, photo-led)	1 Lagoon	2 Reef Night
Photo/video-immersive, premium	1 Lagoon	3 Vivarium
Playful / high-energy	5 Sunset Shore	4 Tiki
Warm, slow, emotional	5 Sunset Shore	3 Vivarium
Data-rich / reference / scholarly	6 Herbarium	7 Spectrum
Developer / maker / dashboard	8 Atelier	7 Spectrum
Professional / business / CRM	8 Atelier	9 Graphite

Default rule: ship both poles and a toggle (the toggle flips tone, not structure). If only one, choose by the app's primary context of use — outdoor/daytime → light; evening/focus/media → dark.

§8 • Quality floor (ship checklist)

- **Responsive:** phone bottom Now-Bar → tablet/desktop left rail; single column → multi-column zones; "needs you" pill → top banner.
- **Contrast:** body text ≥ 4.5:1; large text / UI ≥ 3:1. Don't encode meaning in color alone — pair status with text or icon.
- **Touch targets ≥ 44x44px** (Fitts's).
- **Motion:** < 400 ms interactions; animate transform/opacity; honor `prefers-reduced-motion`.
- **Focus:** visible `:focus-visible` ring on every interactive element.
- **Freshness:** show "updated Xm ago"; treat stale/offline distinctly from bad; media falls back to last image, never a blank panel.
- **No AI-default tells** (§1); one signature element, the rest quiet.
- **Corrections honored:** no "Miller's-7" caps on dashboards; body-doubling framed honestly; no Spectrum graph-paper grid.
- **Version stamp** in three places if the app follows the family convention (HTML comment, `const VERSION`, settings stamp).

§9 • Sources & links

Verified by live web search on 2026-06-13 (per the source research logs in UX_ADHD_RESEARCH.pdf and GROW_CONTINUITY.pdf §11). Canonical UX-law / NN/g URLs are stable; re-verify ADHD links on major revisions.

UX laws & HCI

- Hick's Law — lawsofux.com/hicks-law
- Miller's Law — lawsofux.com/millers-law (+ correction: [UX Myths #23; St  phanie Walter; cobeisfresh](#))
- Von Restorff (isolation) — lawsofux.com/von-restorff-effect
- Fitts's Law — lawsofux.com/fittss-law
- Doherty Threshold — lawsofux.com/doherty-threshold
- Goal-Gradient — lawsofux.com/goal-gradient-effect
- Aesthetic-Usability — lawsofux.com · [NN/g](#)
- Cognitive Load — lawsofux.com/cognitive-load
- Jakob's Law — lawsofux.com/jakobs-law
- Progressive Disclosure — nngroup.com/articles/progressive-disclosure
- Animation duration — nngroup.com/articles/animation-duration
- Cards (UI component) — nngroup.com/articles/cards-component
- Stephen Few, Bullet Graph Design Spec — perceptualedge.com (PDF)

ADHD & app-retention

- R1 dopamine / scaffolding (Lenartowicz & Loo 2014) — brain.fm/blog/focus-apps-for-adhd
- R2 retention failure modes — audhdpsychiatry.co.uk/top-adhd-apps · recallify.ai/adhd-apps-productivity-tools
- R3 body doubling (HCI) — [arXiv 2507.06864](https://arxiv.org/abs/2507.06864); Hart Barnett 2017 — pressbooks.pub
- R4 implementation intentions — Gollwitzer & Sheeran 2006, DOI 10.1016/S0065-2601(06)38002-1 — [full-text PDF \(NIH\)](#)
- R5 time blindness — Barkley 1997 (Psych. Bulletin 121:65–94); Scheres 2006 (Neuropsychologia); understood.org · add.org
- R6 timeboxing / Pomodoro — stimara.com · habi.app

Body-doubling correction

- [Medical News Today](#) (anecdotal, no recent efficacy testing)
- [arXiv 2509.12153](https://arxiv.org/abs/2509.12153) (no conclusive effectiveness; ND discomfort with social presence)
- [Wikipedia: Body doubling](#) (research lacking; mostly master's theses)
- [FLOWN](#) (Virginia Tech thesis: ~17% vs ~12% — modest)
- [ADDA](#) (no research proving effectiveness, but many find it helpful)

\$10 · Optimizations applied + open gaps + changelog

Optimizations baked in (vs the raw exploration)

1. **Token architecture.** Themes became token sets over one shared component layer — the single biggest reuse win; a new app's theme is ~15 variables, not a rebuild.
2. **Two evidence corrections** carried in as rules: body doubling (not ~30%; frame honestly) and Miller's 7±2 (not an on-screen cap).
3. **"When to use cards"** turned into a repeatable rule (§4) after the all-or-nothing miss.
4. **Light/dark = tonal poles of one system** (toggle flips tone, not structure); status colors constant.
5. **Accessibility floor** made explicit (contrast, 44px, focus, reduced-motion, no color-only meaning).
6. **Spectrum grid fix** documented so the graph-paper mistake isn't repeated.
7. **Theme-selection matrix** (§7) so picking is a lookup, not a debate.
8. **Contested effects flagged** (Zeigarnik / endowed-progress) so we don't over-engineer on weak ground.

Open gaps / to-do (next revisions)

- **Per-theme contrast audit** against WCAG for every token pair (esp. `--soft` on `--bg`, and text-over-media) — spot-checked, not yet formally measured.
- **Type scale spec** (a shared modular scale + line-heights) is not yet locked — currently per-screen.
- **Signature components** (Vivarium breathing glow, Herbarium needle gauge, sun-ring, spectrum bar) are described but not yet provided as shared, parameterized code.
- **Source re-verification cadence** — ADHD links last checked 2026-06-13; set a recheck on any MAJOR bump.
- **Motion library** — standardize the snappy easings/durations as named tokens.
- **Density mode** — a compact variant for desktop/data-heavy apps is not yet specified.

Changelog

- **v0.1.0 — 2026-06-20.** Initial UI/UX & theme-system reference. Philosophy, UX-law map, ADHD/retention research with two corrections, card logic, token architecture + shared layer, nine-theme library (Lagoon · Reef Night · Vivarium · Tiki · Sunset Shore · Herbarium · Spectrum · Atelier · Graphite), selection matrix, quality floor, sources. Lagoon light palette locked.